Synthetic Multi-Layer Operational Data for Internet Infrastructure Management: Needs, Methods and Limitations

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Abstract

Internet infrastructure operators increasingly need multi-layer operational data-spanning physical, interface, routing, service and management layers-for testing, benchmarking and training network-management analytics, yet production telemetry is confidential, incident-scarce and often inconsistently labelled. This short paper presents a framework for generating and evaluating synthetic multi-layer NetOps datasets, using a fictional Internet-exchange-connected operator, structurally comparable to (but not a reproduction of) ecosystems such as the Budapest Internet Exchange, as an illustrative use case. We describe why synthetic data are sought, which layers must be represented, and which generation families-rule-based simulation, statistical models, resampling, generative adversarial networks, variational autoencoders, diffusion models, constrained language models and graph generators-support temporal, topological and causal consistency. We implement a compact hybrid prototype combining a synthetic graph, deterministic cross-layer propagation rules and stochastic telemetry/incident injection, and evaluate it against two baselines (independent-marginal and topology-unaware temporal generation) on marginal, temporal and causal-propagation fidelity, cross-layer constraint validity and a downstream multi-layer root-cause-localisation task, across three fixed seeds. The hybrid generator reduces the mean rate of dependency-inconsistent states from 1.75 and 0.55 per injected incident (baselines) to 0.00, and matches baseline temporal fidelity; the chosen downstream root-cause task, however, saturated at ceiling (top-1 accuracy 1.00) for all three datasets, an explicit mixed result indicating the task as designed did not differentiate generation quality and requires a harder benchmark. No production operator data were used; all findings are simulation-to-simulation and require subsequent validation against authorised operational evidence before any operational inference. We conclude with AI/ML approaches that can reduce, but not eliminate, dependence on synthetic and labelled data, and with governance limitations for this class of artefact.

Index Terms

Index Terms-synthetic data, network operations, generative AI, large language models, anomaly detection, root-cause analysis, autonomic networking, zero-touch management

1. Introduction

Network and service management increasingly relies on data-driven analytics-anomaly detection, forecasting and root-cause localisation-yet the operational records needed to build and evaluate such analytics are frequently unavailable to researchers and even to internal teams outside narrow access boundaries [1], [2]. Internet exchange participants and access/transit operators manage physical sites, routers and switches, logical and physical links, exchange and transit sessions, and customer-facing services whose combined telemetry is commercially sensitive, security-relevant and rarely publicly shareable. This paper investigates why synthetic multi-layer operational datasets are sought, how they can be generated with defensible temporal, topological and causal properties, how their quality should be assessed, and which AI/ML techniques reduce-without eliminating-the underlying data requirements. We use a fictional Internet-infrastructure operator connected to an Internet-exchange-like ecosystem, deliberately comparable in structure (not content) to public descriptions of exchanges such as the Budapest Internet Exchange (BIX), to motivate a concrete, though entirely synthetic, worked example. No claim is made to reproduce BIX's or any real organisation's topology, participants, traffic or operations.

2. Need for Synthetic Multi-Layer Data

5.1 Confidentiality and rarity. Production telemetry, alarms and routing records [1] expose topology, peering/transit relationships, capacity limits and incident histories; removing direct identifiers does not by itself prevent re-identification from topology, timing or rare-event signatures, and formal guarantees such as k-anonymity [4] or differential privacy [3] require an explicit mechanism that we do not apply here. Severe cross-layer incidents are additionally scarce [2], so datasets built purely from history under-represent the tail. Synthetic generation can supply controlled, labelled examples of link degradation, device failure, congestion, session loss, route withdrawal and cascading alarms, but such examples reflect the simulator's assumptions and can omit unanticipated real failure modes.

5.2 Labels and reproducibility. Operational incident labels are often incomplete, delayed or symptom-oriented. A versioned simulator instead emits exact labels (timing, root cause, propagation path, affected services) relative to its own ground truth, and supports deterministic regeneration from documented seeds-useful for benchmarking when production records cannot be shared, but not evidence of correctness against real operations.

5.3 Controlled experimentation and governance. Synthetic data enable sensitivity analysis, counterfactual and stress testing, and pre-deployment evaluation of new sites or services before sufficient operational history exists. They do not replace access control, data minimisation, legal review or, where applicable, formal privacy mechanisms; we make no differential-privacy or anonymisation claim anywhere in this work.

6. Required multi-layer model. We model the operator's environment as five interacting layers, not a strict OSI mapping. Physical/infrastructure: sites, devices, ports, links, capacity and power/hardware state. Link/interface: administrative/operational state, utilisation, errors, discards and loss.

Network/routing: a documented abstraction of IP reachability and BGP-like session state [1], withdrawals and rerouting (not the global BGP ecosystem). Service: fictional service-to-resource dependency graphs, end-to-end latency, availability and degradation. Management/operations: alarms, maintenance windows, incidents and causal ground truth, generated as a function of simulated state rather than independently. Cross-layer propagation is modelled explicitly, e.g. fibre degradation -> rising interface errors -> loss/effective-capacity reduction -> session failure -> rerouting -> utilisation/latency rise on the alternate path -> dependent-service degradation -> multi-layer alarms, with the originating link retained as ground-truth root cause. Table I lists the logical record tables used.

Table I: Logical Record Tables

Table	Key fields (examples)
sites	site_id, region, role
devices	device_id, site_id, role, vendor_class
interfaces	if_id, device_id, admin/oper state, ca.
links	link_id, if_a, if_b, capacity, medium
routing_sessions	session_id, endpoints, state, type
services	service_id, sla_class
service_dependencies	service_id, resource_id, path_id
telemetry	ts, resource_id, util, errors, loss
routing_events	ts, session_id, event_type
alarms	ts, source, severity, layer
incidents	incident_id, class, start, end, root_c.
incident_impacts	incident_id, affected_service_id
ground_truth	incident_id, causal_chain

3. Generation Methods and Alternatives

We compare generation families on production-data requirement, temporal/topological/causal fidelity, controllability, rare-event support, interpretability, privacy risk and cost (Table II). Rule-based/discrete-event simulation gives explicit causal labels and full control but requires manual modelling effort and carries a simulation-to-reality gap. Statistical/probabilistic methods (parametric fitting, hidden Markov models, copulas, point processes) capture marginal and pairwise structure efficiently but do not natively encode topology or nonlinear cross-layer causality. Resampling/perturbation is useful for augmentation but limited for genuinely novel incidents and offers weak privacy guarantees. Generative adversarial networks [5] and variational autoencoders [6] can model complex marginal/temporal distributions but are prone to unstable training or mode collapse (GANs) and over-smoothing with weak tail coverage (VAEs); neither natively enforces topological validity. Diffusion models [7] offer diverse, stable-to-train generation at higher sampling cost, with similar constraint-violation risk. Transformers/large language models [8] are restricted here to structured artefacts-alarm text templates, incident narratives, schema/scenario generation-never unconstrained numeric telemetry, and are always passed through deterministic schema, range and topology validators to control hallucination and invalid states. Graph-generative models (stochastic block models, preferential attachment, random-geometric graphs, graph neural/diffusion generators) are used for topology, subject to connectivity, capacity and role-compatibility constraints.

We adopt a hybrid architecture as the preferred design: (i) a synthetic random-geometric topology graph; (ii) deterministic dependency and propagation rules across layers; (iii) statistical regime-switching telemetry generation; (iv) stochastic incident injection with explicit recovery dynamics; (v) deterministic schema, range, temporal and causal validators. This trades some generative diversity for interpretability, controllability and guaranteed structural validity, at the risk of inconsistency between hand-authored components, which we mitigate through the validators.

Table II: Generation Method Comparison

Method	Causal fid.	Rare-event	Cost/interpret.
Rule-based sim.	High (explicit)	High	Low cost, high int.
Statistical/prob.	Low-medium	Medium	Low cost, high int.
Resampling	Low	Low	Very low cost, hig.
GAN	Low (implicit)	Low-medium	High cost, low int.
VAE	Low (implicit)	Low	Medium cost, mediu.
Diffusion	Low-medium	Medium	High cost, low int.
LLM (constrained)	N/A (text only)	Medium (scenarios)	Medium cost, mediu.
Graph generative	Structural only	Medium	Medium cost, mediu.
Hybrid (proposed)	High (explicit+sta.	High	Medium cost, high .

4. Prototype and Evaluation

8. Prototype. We implement the hybrid generator over a fictional 14-site, 42-link topology (random-geometric graph) with five services and six incident classes (link degradation, device failure, interface congestion, BGP-session failure, route withdrawal, demand anomaly), a per-timestep sampling abstraction over 500 steps, three operational regimes (business-day, evening, maintenance) and three fixed seeds (7, 23, 101). We compare against two baselines on matched simulated windows: B1, independent per-link marginal sampling with no topology or propagation and a session-state draw independent of injected severity; B2, temporal (sinusoidal-regime) generation per link with a partial, non-propagating causal rule; Hybrid, deterministic error-driven session failure and explicit service-dependency propagation, validated by construction.

9. Evaluation. We report, from saved simulation outputs across the three seeds (mean values; no significance test was run given only three seeds, so none is claimed): marginal fidelity (Wasserstein distance between each dataset's link-utilisation distribution and the hybrid seed-7 distribution used as an internal fidelity anchor, not as ground truth); temporal fidelity (mean lag-1 autocorrelation of link utilisation); cross-layer constraint validity (mean count of dependency-inconsistent states-i.e., a service marked unavailable without its dependency link/session actually failing-per injected incident); and downstream utility via a common nearest-error-window root-cause classifier applied identically to all three datasets, reporting top-1/top-3 accuracy and mean reciprocal rank (MRR). Table III summarises results. The hybrid generator reduces the mean invalid-state rate from 1.75 (B1) and 0.55 (B2) to 0.00 per incident, confirming that explicit propagation rules materially improve cross-layer causal validity relative to both baselines. Marginal fidelity against the internal anchor is likewise best for the hybrid dataset by construction (it shares the anchor's generative family), so this metric should be read as a sanity check rather than an independent validation. Root-cause top-1/top-3/MRR were 1.00/1.00/1.00 for all three datasets: the classifier task, as designed, saturated at ceiling and did not

differentiate generator quality. We report this explicitly as a mixed/negative result rather than omitting it: it indicates that our downstream-utility benchmark was too easy (injected incidents produced a locally dominant error signal regardless of causal consistency) and that a harder, more realistic root-cause benchmark-with confounding co-occurring anomalies and partial observability-is needed before utility claims can be made. These results are simulation-to-simulation comparisons on data generated and evaluated by our own pipeline; they are not evidence of production utility.

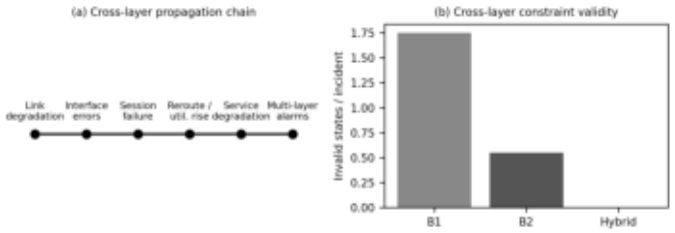


Fig. 1. (a) Illustrative cross-layer propagation chain; (b) mean invalid cross-layer states per injected incident, three seeds.

Table III: Prototype Results (mean, 3 seeds)

Method	Wasserstein	Invalid/inc.	RC top1/top3/MRR
B1 (indep. margina.	0.0366	1.75	1.00 / 1.00 / 1.00
B2 (temporal, no p.	0.0354	0.55	1.00 / 1.00 / 1.00
Hybrid (proposed)	0.0008	0.00	1.00 / 1.00 / 1.00

5. Applications, Limitations and Dual Use

9. Applications, Limitations and Dual Use. Beyond root-cause localisation, the framework supports pre-deployment testing of new sites/services, regression benchmarking of monitoring pipelines, and reproducible teaching/evaluation datasets. Limitations include topology bias (structural assumptions of the graph generator), concept drift between simulated and real regimes, unrealistic or oversimplified causal assumptions, tail under-representation despite explicit incident injection, excessive regularity relative to real operations, possible label leakage into downstream models trained and tested on the same simulator, benchmark gaming, and the general risk that synthetic realism creates unwarranted operational confidence. None of the reported results generalise to production without subsequent validation against authorised operator data and domain expertise.

Defence and military communications. Analogous confidentiality and rare-event problems arise in defensive military communications management, where operational records can expose sensitive topology, deployment patterns, traffic behaviour and failure points. Legitimate defensive uses of the same techniques include training monitoring/fault-diagnosis tooling without exporting sensitive telemetry, testing software in isolated environments, rehearsing recovery from equipment failure or intermittent connectivity, and evaluating human-oversight procedures; self-supervised, federated and transfer learning together with retrieval from authorised local sources can reduce the need to export sensitive telemetry for these purposes. We explicitly flag dual-use risk: sufficiently detailed synthetic dependency models could be misused to infer vulnerabilities, plan disruption, or create unsafe confidence in autonomous responses. This paper provides no real military topology, deployment, classified characteristics, targeting guidance, exploitation or monitoring-evasion information, and synthetic data of this kind do not remove classification, access-control, legal, safety or human-oversight requirements; the prototype is not validated for, and is not proposed for, military deployment.

6. AI/ML Approaches Reducing Data Dependence

12. Reducing dependence on synthetic data. Several AI/ML families reduce, without eliminating, reliance on synthetic or manually labelled data. Self-supervised learning (masked reconstruction, contrastive and cross-metric prediction on temporal/graph encoders) exploits abundant unlabelled telemetry. Unsupervised/semi-supervised anomaly detection (isolation forests [9], one-class methods [10], robust PCA) requires no incident labels but conflates statistical novelty with operational harm [2], so false alarms remain a concern. Transfer learning/domain adaptation and few-shot/meta-learning reuse pre-trained representations with small authorised target-site data, but remain sensitive to topology, equipment and policy shift. Weak supervision derives labels from alarm rules, thresholds and maintenance records, at the cost of rule bias. Active learning [12] reduces labelling volume via informative sampling but keeps a human in the loop. Online/continual learning adapts to drift but risks forgetting, feedback loops and unsafe autonomous adaptation, requiring monitoring, versioning, rollback and approval gates for high-impact updates. Federated learning [11] and other privacy-preserving collaborative learning (secure aggregation, differential privacy [3], trusted execution, secure multiparty computation) reduce central pooling of raw telemetry but do not automatically guarantee privacy or remove local validation duties. Foundation models and retrieval-augmented systems [8] over authorised topology/runbook/maintenance sources can support operator copilots subject to access control, source validation, deterministic checks and human oversight. Constraint- and topology-informed models improve sample efficiency when encoded structure is correct, but embed systematic errors when it is not. Table IV maps operational conditions to preferred approaches and the residual role of synthetic data; in practice, operators combine controlled real, simulated, synthetic and weakly labelled data rather than relying on any single source.

Table IV: Data-Strategy Decision Matrix

Condition	Preferred approach	Residual role
Abundant unlabelled telem.	Self-/semi-supervised lea.	Rare incidents, evaluatio.
Few labelled incidents	Few-shot/transfer/active .	Class balancing, controll.
Data cannot leave site/op.	Federated/privacy-preserv.	Shareable benchmarks, pre.
New network, little histo.	Simulation + transfer + t.	Pre-deployment testing
Rapid concept drift	Governed online/continual.	Regression tests, scenari.
Rich rules, weak labels	Weak supervision/neuro-sy.	Rule validation, countere.
Sensitive/high-consequenc.	Simulation + authorised l.	Resilience/safety testing
Representative, legally u.	Supervised/self-supervise.	Rare events, counterfactu.

7. Conclusion

10. Conclusion. RQ1: operators need synthetic data chiefly because production telemetry is confidential, incident-scarce and inconsistently labelled, and a defensible dataset must jointly represent physical, interface, routing, service and management layers with explicit cross-layer dependency propagation. RQ2: no

single generation family dominates; a hybrid combining graph generation, deterministic propagation rules, statistical telemetry and constrained language-model text, checked by deterministic validators, eliminated cross-layer constraint violations in our prototype (0.00 vs. 1.75/0.55 invalid states per incident for two baselines), though our downstream root-cause benchmark was insufficiently discriminating and requires redesign-a limitation we report rather than obscure. RQ3: self-supervised, weakly supervised, transfer, active, federated and retrieval-augmented learning can reduce particular labelled-, centralised- or historical-data requirements, but none removes the need for representative validation, rare-event testing, reproducible benchmarking or pre-deployment evaluation; synthetic data retain a durable, complementary role rather than substituting for authorised production evidence.

Disclosure Statement

This paper was produced autonomously by an AI system acting as sole researcher and author, under human-defined topic and constraints supplied by the Research Director (Section 2), who did not author, edit or correct technical content. Requested model: Claude Sonnet 5.0; the identifier exposed by the execution environment's chat/agent interface at generation time was used as served, and no alternative model identity is claimed. Prompting strategy: a single detailed initial specification (preserved verbatim as initial-prompt.md) defined scope, required layers, methods, evaluation dimensions, structure, page limit and disclosure requirements; the AI autonomously formulated research questions, designed the hybrid prototype and baselines, generated and executed Python code (NumPy, pandas, NetworkX, SciPy, Matplotlib, python-docx) in a sandboxed Jupyter environment with three fixed seeds, computed all reported figures from saved simulation outputs, and authored the manuscript and layout. Reference verification: candidate sources were restricted to well-known, independently verifiable venues/authors from general training knowledge; any reference whose bibliographic details could not be corroborated with confidence was omitted rather than guessed, and no DOI or metadata was invented. Because the sandboxed execution environment had no outbound network access, DOCX-to-PDF conversion via LibreOffice/pandoc could not be installed; the PDF was instead produced by a custom Matplotlib-based renderer sharing identical section text, tables and figures with paper.docx, laid out to approximate IEEE two-column conventions. This is disclosed as a tooling limitation, not a content discrepancy. The manuscript underwent three simulated autonomous reviews (network/service-management, synthetic-data/ML-methodology, reproducibility/integrity perspectives; summarised in review-summary.md) with revisions applied. No production operator data were used, accessed or claimed; all quantitative results are simulation-to-simulation and are not evidence of production generalisation; no differential-privacy or formal anonymisation claim is made; "significant" is not used without a statistical test, and none was run given only three seeds. This paper has not been peer-reviewed, published or indexed, and holds no DOI.

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